

CHAPTER 5

I'm Up and Down, and Round and Round

NCERT Exercise Solutions

Introduction	3 Q
Think and Reflect · 5.2 Symmetries of a Circle	3 Q
Think and Reflect · 5.3 How Many Circles?	6 Q
Exercise Set 5.1	4 Q
Think, Draw and Infer	2 Q
Exercise Set 5.2	2 Q
Exercise Set 5.3	3 Q
5.6 Distance of Chords from the Centre (Activity)	1 Q
Exercise Set 5.4	3 Q
5.6.1 Which of the two unequal chords is farther from the centre? (Activity)	1 Q
Exercise Set 5.5	3 Q

✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.
Independent study aid by Beginners Point. Not an official NCERT publication.

⭐ Introduction

? **Can you recognise the origin of the shapes in Fig. 5.1?**

> 💡 **SAMPLE ANSWER**
>
> The circles, ovals and other shapes were likely inspired by things in nature: the Sun and full Moon (circles), ripples when raindrops fall on water, cross-sections of plant stems, flowers like the sunflower, eggs and leaves (ovals). (check the picture in your book)

Activity **List some objects from nature that resemble a circle.**

> 💡 **SAMPLE ANSWER**
>
> - Full Moon and the Sun
> - Ripples on water when a raindrop or stone falls
> - Cross-section of a tree trunk or plant stem
> - Sunflower head
> - Pupil and iris of the eye
> - A spider web (roughly)

Think and Reflect **Jamuna has a circular piece of paper. She is trying to locate its centre. Amina gives her a suggestion. She follows the instructions and is thrilled to find**

that it works. Can you guess what Amina told her?

💡 SAMPLE ANSWER

Amina probably said:

- Fold the paper exactly in half so the edges overlap. The crease is a diameter.
- Open it and fold in half again in a different direction. This crease is another diameter.
- **The point where the two creases cross is the centre**, because every diameter passes through the centre.

🤔 Think and Reflect · 5.2 Symmetries of a Circle

1 **What are the rotational symmetries of a square? How many lines of reflection symmetry does it have? What about a regular pentagon? A regular hexagon?**

✔ ANSWER

Shape	Rotational symmetries (angles)	Lines of reflection
Square	90°, 180°, 270°, 360° (4)	4
Regular pentagon	72°, 144°, 216°, 288°, 360° (5)	5
Regular hexagon	60°, 120°, 180°, 240°, 300°, 360° (6)	6

A regular polygon with n sides has **n rotational symmetries** (multiples of 360° ÷ n) and **n lines of reflection symmetry**.

2 **What is the length of the longest chord in a circle of radius 5 units? Is there a smallest chord?**

✔ ANSWER

The longest chord is the diameter = 2 × 5 = **10 units**.

There is no smallest chord. As a chord moves away from the centre it gets shorter and shorter; for any chord we can find a still shorter one. Its length gets as close to 0 as we like.

3 **The locus of points at a given distance from a given point is a circle. What can we say about the locus of points equidistant from two given points A and B?**

✔ ANSWER

Let M be the midpoint of AB and P any point on the perpendicular bisector of AB.

In ΔPMA and ΔPMB: AM = BM, ∠PMA = ∠PMB = 90°, PM is common.

So ΔPMA ≅ ΔPMB (SAS), hence PA = PB.

So every point on the perpendicular bisector is equidistant from A and B, and every point equidistant from A and B lies on it.

The locus is the perpendicular bisector of AB.

🤔 Think and Reflect · 5.3 How Many Circles?

1 **How many circles pass through two points on a plane?**

✔ ANSWER

Every point on the perpendicular bisector of AB can be a centre.

So **infinitely many circles** pass through two points.

2 **Are there circles of all possible radii passing through A and B? What is the radius of the smallest circle passing through A and B? What is the radius of the largest circle passing through A and B?**

✔ ANSWER

No. The radius can never be less than half of AB (the chord AB can't be longer than the diameter).

Smallest circle: centre at the midpoint of AB, **radius = AB/2** (AB is a diameter).

There is no largest circle: moving the centre farther along the perpendicular bisector makes the radius as large as we like.

3 **As you move away from segment AB along its perpendicular bisector, do the radii of the circles containing A and B increase or decrease?**

✔ ANSWER

If the centre is at distance d from the midpoint M, radius = $\sqrt{((AB/2)^2 + d^2)}$.

As d increases, **the radius increases**.

4 **As you go along the perpendicular bisector, will the circle drawn from that point through A and B appear more curved or less curved?**

✔ ANSWER

The circles get bigger, so they appear **less curved (flatter)**. Very far away, the arc near AB looks almost like a straight line.

5 **You are given two points A and B on a plane. How many squares can you draw with A and B on the boundary? How many squares can you draw with A and B as the corners of the square?**

✔ ANSWER

With A and B just on the boundary: **infinitely many squares** (squares of many sizes and positions can have A and B on their sides).

With A and B as corners: **3 squares**:

- AB as a side, square on one side of AB

- AB as a side, square on the other side of AB
- AB as a diagonal (one square)

? **If A, B and C are collinear, can you explain why there is no circle through A, B and C?**

✔ ANSWER

The centre would have to be equidistant from A, B and C, so it must lie on the perpendicular bisectors of AB and of BC.

Both bisectors are perpendicular to the same line ABC, so they are parallel and never meet.

So no such centre exists, and **no circle passes through three collinear points**.

📝 Exercise Set 5.1

1 **Draw ΔABC with AB = 5 cm, ∠A = 70° and ∠B = 60°. Draw the circumcircle of ΔABC. Is the centre inside or outside the triangle?**

✔ ANSWER

Steps:

- Draw AB = 5 cm. At A make 70°, at B make 60°; the rays meet at C.
- Draw the perpendicular bisectors of AB and BC (compass arcs). They meet at O.
- With centre O and radius OA, draw the circle. It passes through A, B, C.

$\angle C = 180° - 70° - 60° = 50°$. All three angles are acute.

So the centre is **inside the triangle** (radius ≈ 3.3 cm).

2 **Draw ΔABC with AB = 5 cm, ∠A = 100°, AC = 4 cm. Draw the circumcircle of ΔABC. Is the centre inside or outside the triangle?**

✔ ANSWER

Steps: draw AB = 5 cm, make 100° at A, mark C on that ray with AC = 4 cm, join BC.

Draw perpendicular bisectors of AB and AC; they meet at O. Draw the circle with centre O, radius OA.

∠A = 100° is obtuse, so the centre is **outside the triangle** (beyond side BC; radius ≈ 3.5 cm).

3 **Draw ΔABC with AB = 6 cm, BC = 7 cm and CA = 7 cm. Draw the circumcircle of ΔABC. Let the circumcentre be O. Measure OA, OB, OC.**

✔ ANSWER

Draw AB = 6 cm; with A and B as centres and radius 7 cm, draw arcs meeting at C.

Draw perpendicular bisectors of two sides; they meet at O (inside, since the triangle is acute). Draw the circle.

OA, OB and OC are all radii, so they are equal.

Check by calculation: height from C = $\sqrt{(7^2 - 3^2)} = \sqrt{40} \approx 6.32$ cm; area $\approx \frac{1}{2} \times 6 \times 6.32 \approx 18.97$ cm^2; R = $(6 \times 7 \times 7) \div (4 \times 18.97) \approx 3.87$ cm.

OA = OB = OC ≈ 3.9 cm.

4 **What is the least possible radius of a circle through two points A and B?**

✔ ANSWER

The smallest circle has AB as diameter, centre at the midpoint of AB.

Least radius = AB/2 (half the length of AB).

🤔 Think, Draw and Infer

1 **A, B and C are three collinear points. Can you find a point P such that PA = PB = PC? What can you say about the perpendicular bisectors of AB and BC? Draw and check.**

Can you show that for three collinear points A, B and C, the perpendicular bisectors of AB and BC are parallel? Is it possible for a circle to pass through collinear points? Can you draw a line that cuts a given circle in three distinct points?

✔ ANSWER

The perpendicular bisectors of AB and BC are both perpendicular to the same line ABC.

Two lines perpendicular to the same line are parallel (they pass through different midpoints, so they are not the same line).

So they never meet: **there is no point P with PA = PB = PC.**

Hence **no circle passes through three collinear points.**

So **no line can cut a circle in three distinct points**; a line meets a circle in at most 2 points.

2 **The circumcircle of a given ΔABC is drawn. Can there be other triangles congruent to ΔABC that share the same circumcircle?**

✔ ANSWER

Yes, infinitely many. Rotate ΔABC about the centre O by any angle: the vertices stay on the circle and the new triangle is congruent to ΔABC.

Reflecting ΔABC in any diameter also gives a congruent triangle on the same circle.

📝 Exercise Set 5.2

1 **Show that the triangle formed by a chord and the centre of the circle is isosceles.**

✔ ANSWER

Let AB be a chord of a circle with centre O.

OA = OB = r (both are radii).

So ΔOAB has two equal sides: **it is isosceles**.

2 **Show that if two such isosceles triangles (occurring in the previous question) have equal base length, they are congruent to each other.**

✔ ANSWER

Let AB and DE be chords of a circle with centre O, AB = DE.

In ΔOAB and ΔODE:

- OA = OD (radii)
- OB = OE (radii)
- AB = DE (given)

So **ΔOAB ≅ ΔODE** (SSS).

📝 Exercise Set 5.3

1 **Can you explain why the converse of Theorem 4 is true, i.e., why does the perpendicular from the centre of a circle to a chord of the circle bisect the chord? (Use Fig. 5.12: ∠CMA = ∠CMB = 90°.)**

✔ ANSWER

In ΔCMA and ΔCMB:

- ∠CMA = ∠CMB = 90°
- CA = CB (radii, hypotenuses)
- CM is common

So ΔCMA ≅ ΔCMB (RHS).

Hence **AM = BM**: the perpendicular bisects the chord.

2 **An isosceles triangle ABC is inscribed in a circle, with AB = AC. Show that the altitude from A to BC passes through the centre of the circle.**

✔ ANSWER

Let AD be the altitude from A to BC (D on BC).

In ΔADB and ΔADC: AB = AC, AD common, ∠ADB = ∠ADC = 90°. So ΔADB ≅ ΔADC (RHS) and BD = DC.

So AD is perpendicular to BC and passes through its midpoint: **AD is the perpendicular bisector of chord BC.**

The centre O is equidistant from B and C (OB = OC = r), so O lies on the perpendicular bisector of BC.

Hence **the altitude from A passes through the centre.**

3 **Two parallel chords of lengths 6 cm and 8 cm are on opposite sides of the centre of a circle. If the radius of the circle is 5 cm, find the distance between the midpoints of the chords.**

✔ ANSWER

The perpendicular from the centre O bisects each chord; as the chords are parallel, both perpendiculars lie on one line through O.

Chord 6 cm: half = 3, distance = $\sqrt{(5^2 - 3^2)} = \sqrt{16} = 4$ cm.

Chord 8 cm: half = 4, distance = $\sqrt{(5^2 - 4^2)} = \sqrt{9} = 3$ cm.

Opposite sides of the centre: distance = 4 + 3 = **7 cm**.

5.6 Distance of Chords from the Centre (Activity)

? Now draw another chord of the same length. How will you do this? Join the centre to the midpoint of the new chord and measure its length. Is it the same as the distance from the centre to the first chord?

✔ ANSWER

Measure the first chord with the compass. Put the compass point anywhere on the circle and cut the circle with an arc; join the two points. This chord has the same length.

On measuring, the distance from the centre to its midpoint is **the same** as for the first chord.

Exercise Set 5.4

1 Use the Baudhāyana–Pythagoras theorem to show why Theorem 6 must be true. (Fig. 5.14:

AB = FG, E and H are midpoints, C the centre.)

✔ ANSWER

CE ⊥ AB and CH ⊥ FG; AE = AB/2, FH = FG/2, CA = CF = r.

In right ΔCEA: $CE^2 = CA^2 - AE^2 = r^2 - (AB/2)^2$.

In right ΔCHF: $CH^2 = CF^2 - FH^2 = r^2 - (FG/2)^2$.

AB = FG, so $CE^2 = CH^2$, hence **CE = CH**: equal chords are equidistant from the centre.

2 **Consider Fig. 5.15. If CE is perpendicular to AB, CH is perpendicular to GF, and CE = CH, show that AB = GF.**

✔ ANSWER

In ΔCEA and ΔCHG:

- ∠CEA = ∠CHG = 90°
- CA = CG (radii)
- CE = CH (given)

So ΔCEA ≅ ΔCHG (RHS) and AE = GH.

By Theorem 5, E and H are midpoints of AB and GF. So AB = 2AE = 2GH = GF.

Hence **AB = GF**.

3 **Solve the previous question using the Baudhāyana–Pythagoras theorem.**

✔ ANSWER

Let CE = CH = d and radius = r.

$AE^2 = CA^2 - CE^2 = r^2 - d^2$.

$GH^2 = CG^2 - CH^2 = r^2 - d^2$.

So AE = GH. E and H are midpoints, so AB = 2AE = 2GH = GF. **AB = GF.**

5.6.1 Which of the two unequal chords is farther from the centre? (Activity)

? **You have two chords on a circle; one is longer than the other. Which chord is closer to the centre? Can you guess?**

Activity: Draw chords of various lengths, drop perpendiculars from the centre and record the length and distance in Table 1. What do you observe?

💡 **SAMPLE ANSWER**

Example: circle of radius 5 cm (distance = √(25 − (half-chord)²)):

Length of chord	10 cm	8 cm	6 cm	4 cm
Distance from centre	0 cm	3 cm	4 cm	≈ 4.6 cm

The longer the chord, the closer it is to the centre (the diameter is the longest chord, at distance 0).

Exercise Set 5.5

1 **Find the length of the chord of a circle where the radius is 7 cm and perpendicular distance**

is 6 cm.

✔ ANSWER

Half chord = $\sqrt{(7^2 - 6^2)} = \sqrt{(49 - 36)} = \sqrt{13}$ cm.

Chord = $2\sqrt{13}$ cm ≈ **7.21 cm** (**$2\sqrt{13}$ cm**).

2 **Explain why: if the perpendicular distance of a chord from the centre is d and the radius is r, then the chord length is $2\sqrt{(r^2 - d^2)}$.**

✔ ANSWER

Let chord AB, centre O, and OM ⊥ AB with OM = d.

By Theorem 5, M is the midpoint, so AB = 2AM.

In right ΔOMA: $AM^2 = OA^2 - OM^2 = r^2 - d^2$, so $AM = \sqrt{(r^2 - d^2)}$.

Hence **$AB = 2\sqrt{(r^2 - d^2)}$**.

3* **In a circle, if the distance of chord AB from the centre is twice the distance of another chord CD from the centre, then can we conclude that CD = 2AB? Give reasons.**

✔ ANSWER

No. Let CD be at distance d and AB at distance 2d.

CD = $2\sqrt{(r^2 - d^2)}$ and AB = $2\sqrt{(r^2 - 4d^2)}$.

Example: r = 5, d = 1: CD = $2\sqrt{24}$ ≈ 9.8, AB = $2\sqrt{21}$ ≈ 9.2. CD is not 2AB.

CD = 2AB would need $r^2 - d^2 = 4(r^2 - 4d^2)$, i.e. $r^2 = 5d^2$, which happens only in that special case.

The chord length is not proportional to its distance from the centre.

⭐ 5.7 Angles Subtended by an Arc

Exercise **A circle with centre O has points A, B, C, D on it (Fig. 5.19). Measure the angles subtended by arc AKB and arc CLD at the centre O. State whether arcs AKB and CLD are minor or major arcs.**

💡 SAMPLE ANSWER

Arc AKB (on the left, between A and B via K): angle at O is about **100°**, less than 180° → **minor arc**.

Arc CLD (from C to D via L on the right side): angle at O is about **200°**, more than 180° → **major arc**.

(Measured values may differ a little; check the picture in your book.)

Activity **Fix an arc AKB. Measure the angle it subtends at the centre and at three points P, Q, R on the circle outside the arc. What do you notice?**

💡 SAMPLE ANSWER

∠APB = ∠AQB = ∠ARB: the angles at P, Q, R are all equal.

Each is **half the angle subtended at the centre** (e.g. centre angle 90° → 45° at P, Q, R).

📝 Exercise Set 5.6

1 **In a circle with centre O, the central angle AOB is 60°. If the radius of the circle is 12 cm, what is the length of the chord AB?**

✔ ANSWER

OA = OB = 12 cm, so ∠OAB = ∠OBA = (180° − 60°) ÷ 2 = 60°.

All angles are 60°, so ΔOAB is equilateral.

AB = 12 cm.

2 (i) **Let A and B be two points on a circle with centre O. Are there points X, Y on the circle, on the same side of AB, such that ∠AXB is different from ∠AYB?**

✔ ANSWER

No. X and Y on the same side of AB lie on the same arc, outside the other arc AB.

Each angle is half the angle that arc subtends at the centre, so ∠AXB = ∠AYB (angles in the same segment are equal).

2 (ii) **Is it true that if ∠AXB = ∠AYB, then X and Y lie on the same side (of AB)?**

✔ ANSWER

Not always. If X and Y are on opposite sides, AXBY is a cyclic quadrilateral, so ∠AXB + ∠AYB = 180°.

They can be equal only when both are 90°, i.e., when AB is a diameter.

So it is true **except when AB is a diameter** (then X, Y can be on opposite sides).

2 (iii) **If ∠AXB = ∠AYB, and X and Y do not lie on the circle, does the circle through A, B and X also pass through Y?**

✔ ANSWER

Yes, if X and Y are on the same side of AB. By Theorem 10, segment AB subtends equal angles at X and Y on the same side, so A, B, X, Y are concyclic: the circle through A, B, X passes through Y.

If X and Y are on opposite sides of AB, it need not pass through Y.

3 **Find x in Fig. 5.26.**
(A, D, C, B lie on a circle; ∠ADC = 100° at D; x = ∠ABC at B, opposite to D.)

✔ ANSWER

ABCD is a cyclic quadrilateral; D and B are opposite vertices.

$x + 100° = 180°$

$x = 80°$

⭐ 5.8 Concyclicity of Points

Exercise **A cyclic quadrilateral has angles $\angle A = 80°$, $\angle B = 110°$, $\angle C = 100°$, and $\angle D = 70°$. Can such a quadrilateral be drawn? Explain why or why not.**

✔ ANSWER

Sum = $80° + 110° + 100° + 70° = 360°$ ✓

$\angle A + \angle C = 80° + 100° = 180°$ and $\angle B + \angle D = 110° + 70° = 180°$ ✓

Opposite angles are supplementary, so by Theorem 12 such a quadrilateral is cyclic. **Yes, it can be drawn.**

? **Theorem 12: we dealt only with the case where C is outside the circle through A, B, D. Complete the other case (C inside the circle).**

✔ ANSWER

Suppose C is inside the circle through A, B, D. Extend DC beyond C to meet the circle at E.

ABED is cyclic, so $\angle BAD + \angle BED = 180°$ (Theorem 11).

Given ∠BAD + ∠BCD = 180°. So ∠BCD = ∠BED.

But D, C, E are on a line, so ∠BCD is the exterior angle of ΔBCE at C, hence ∠BCD > ∠BEC = ∠BED.

This is a contradiction. So C cannot be inside; **A, B, C, D lie on a circle.**

End-of-Chapter Exercises

1 **In a circle, a chord is 5 cm away from the centre. If the radius of the circle is 13 cm, what is the length of the chord?**

✔ ANSWER

Half chord = $\sqrt{(13^2 - 5^2)} = \sqrt{(169 - 25)} = \sqrt{144} =$ 12 cm.

Chord = 2 × 12 = **24 cm**.

2 **An arc of a circle subtends an angle of 70° at the centre. What is the measure of the angle subtended by the arc at a point on the circle (outside the arc)?**

✔ ANSWER

Angle at the circle = ½ × angle at the centre = ½ × 70° = **35°**.

3 **The diameter of a circle is 26 cm. A chord of length 24 cm is drawn in the circle. Find the**

distance from the centre of the circle to the chord.

✔ ANSWER

r = 26 ÷ 2 = 13 cm; half chord = 12 cm.

Distance = $\sqrt{(13^2 - 12^2)} = \sqrt{(169 - 144)} = \sqrt{25} =$ **5 cm**.

4 **A circle has a radius of 15 cm. The distance from the centre of the circle to a chord is 9 cm. What is the length of the chord?**

✔ ANSWER

Half chord = $\sqrt{(15^2 - 9^2)} = \sqrt{(225 - 81)} = \sqrt{144} =$ 12 cm.

Chord = **24 cm**.

5 **Prove that the perpendicular bisector of a chord passes through the centre of the circle.**

✔ ANSWER

Let AB be a chord of a circle with centre O.

OA = OB (radii), so O is equidistant from A and B.

The locus of points equidistant from A and B is the perpendicular bisector of AB.

So O lies on the perpendicular bisector: **the perpendicular bisector of a chord passes through the centre.**

6 **The diameter of a circle is AB. Point C is on the circumference. What is the measure of ∠ACB? Explain your reasoning.**

✔ ANSWER

The arc AB not containing C subtends a straight angle ∠AOB = 180° at the centre O.

∠ACB = ½ × 180° = **90°** (angle in a semicircle).

7 **ABCD is a cyclic quadrilateral. If ∠A measures 75°, what is the measure of ∠C? If ∠B measures 110°, what is the measure of ∠D?**

✔ ANSWER

Opposite angles of a cyclic quadrilateral add up to 180°.

∠C = 180° − 75° = **105°**

∠D = 180° − 110° = **70°**

8 **Quadrilateral PQRS is inscribed in a circle. If ∠P = (2x + 10)° and ∠R = (3x − 20)°, find the value of x and the measures of ∠P and ∠R.**

✔ ANSWER

∠P + ∠R = 180°

2x + 10 + 3x − 20 = 180

5x − 10 = 180 → 5x = 190 → **x = 38**

$\angle P = 2 \times 38 + 10 = \mathbf{86°}$; $\angle R = 3 \times 38 - 20 = \mathbf{94°}$ (check: 86 + 94 = 180 ✓)

9 **The distance of a chord of length 16 cm from the centre of a circle is 6 cm. Find the radius of the circle.**

✔ ANSWER

Half chord = 8 cm.

$r = \sqrt{(8^2 + 6^2)} = \sqrt{(64 + 36)} = \sqrt{100} = \mathbf{10\ cm}$.

10 **A cyclic quadrilateral has sides 5, 5, 12, 12 units. Find its area.**

✔ ANSWER

Case 1: sides in order 5, 12, 5, 12 → a parallelogram; a cyclic parallelogram is a rectangle. Area = 5 × 12 = 60.

Case 2: sides in order 5, 5, 12, 12 → a kite ABCD with AB = AD = 5, CB = CD = 12. $\Delta ABC \cong \Delta ADC$ (SSS), so $\angle B = \angle D$; as $\angle B + \angle D = 180°$, $\angle B = \angle D = 90°$.

Area = 2 × (½ × 5 × 12) = 60.

Area = 60 square units (either way).

11* **Consider a cyclic quadrilateral. Without drawing its circumcircle, how can we find out whether the centre of the circumcircle lies inside the quadrilateral or outside? What is the best way of finding out?**

💡 **SAMPLE ANSWER**

Look at the angle each side subtends at the opposite vertices, i.e. the angle between a side and a diagonal: ∠ACB (for side AB), ∠BDC (for BC), ∠CAD (for CD), ∠DBA (for DA).

- If a side subtends an **obtuse** angle, the centre lies beyond that side → **outside** the quadrilateral.
- If it subtends exactly 90°, that side is a diameter → centre **on** that side.
- If all four such angles are **acute** → centre **inside**.

(Reason: a chord subtends an acute angle at points on the same side as the centre, and an obtuse angle at points on the other side.)

12* **When two chords intersect, each is divided into two line segments. Show that if the intersecting chords are of equal length, then the segments of one chord are equal to the corresponding segments of the other.**

✔ **ANSWER**

Let equal chords AB and CD meet at P. Draw $OM \perp AB$ and $ON \perp CD$ (O the centre). Label so that A and C are on the same side as in the figure (P between M and B and between N and D).

Equal chords are equidistant from the centre: OM = ON.

$\Delta OMP \cong \Delta ONP$ (RHS: OM = ON, OP common, right angles at M, N). So MP = NP.

M, N are midpoints, so AM = AB/2 = CD/2 = CN.

AP = AM + MP = CN + NP = CP.

PB = AB − AP = CD − CP = PD.

So **AP = CP and PB = PD**. (If P is the midpoint of one chord, then M = N = P and both are bisected.)

13* **Draw a circle in which a chord of 6 cm length stands at a distance of 3 cm from the centre. (Hint: Is it a circumcircle of a suitable triangle?)**

✔ ANSWER

Steps:

- Draw AB = 6 cm and its perpendicular bisector; let M be the midpoint.
- On the bisector mark O with MO = 3 cm.
- With centre O and radius OA draw the circle.

Radius = $\sqrt{(3^2 + 3^2)} = \sqrt{18} = 3\sqrt{2} \approx$ **4.24 cm**.

(Hint: the circle is the circumcircle of ΔOAB or of any triangle ABX on it; note ∠AOB = 90°.)

14* **Show that a rectangle is the only parallelogram that can be inscribed in a circle.**

✔ ANSWER

Let parallelogram ABCD be inscribed in a circle (cyclic).

Cyclic: ∠A + ∠C = 180°.

Parallelogram: ∠A = ∠C (opposite angles equal).

So 2∠A = 180°, ∠A = 90°: ABCD is a **rectangle**.

And every rectangle can be inscribed (opposite angles 90° + 90° = 180°, Theorem 12).

15* **Show that if a rectangle is inscribed in a circle, then the point of intersection of its diagonals must lie at the centre of the circle.**

✔ ANSWER

Let rectangle ABCD be inscribed; diagonals meet at P.

Diagonals of a rectangle are equal and bisect each other, so PA = PB = PC = PD.

P is equidistant from three non-collinear points A, B, C on the circle. By Theorem 1 only one point (the circumcentre) has this property.

So **P is the centre of the circle.**

(Or: ∠ABC = 90°, so AC subtends 180° at the centre and is a diameter; likewise BD. Two diameters meet at the centre.)

16* **Consider all chords of a circle of a fixed length. What is the shape formed by the midpoints of all these chords?**

✔ ANSWER

Chords of length L in a circle of radius r are all at distance $d = \sqrt{(r^2 - (L/2)^2)}$ from the centre, and this distance is from the centre to the midpoint.

So all midpoints are at the same distance d from the centre.

They form a circle with the same centre and radius $\sqrt{(r^2 - L^2/4)}$ (only the centre itself if the chords are diameters).

17* **In a circle with centre O, chords AB and AC are congruent. Explain why "the centre of the circle lies on the angle bisector of ∠BAC".**

✔ ANSWER

Join OA, OB, OC.

In ΔOAB and ΔOAC: OB = OC (radii), AB = AC (given), OA common.

So ΔOAB ≅ ΔOAC (SSS) and ∠OAB = ∠OAC.

So AO divides ∠BAC into two equal parts: **O lies on the bisector of ∠BAC.**

18 **Two parallel chords of lengths 10 cm and 24 cm are on the same side of the centre of a circle. The distance between the chords is 7 cm. Find the radius of the circle.**

✔ ANSWER

Let the radius be r. Distances from the centre: 10 cm chord: $\sqrt{(r^2 - 25)}$; 24 cm chord: $\sqrt{(r^2 - 144)}$.

Same side: $\sqrt{(r^2 - 25)} - \sqrt{(r^2 - 144)} = 7$.

$\sqrt{(r^2 - 25)} = 7 + \sqrt{(r^2 - 144)}$. Square: $r^2 - 25 = 49 + 14\sqrt{(r^2 - 144)} + r^2 - 144$

$70 = 14\sqrt{(r^2 - 144)} \rightarrow \sqrt{(r^2 - 144)} = 5 \rightarrow r^2 = 169$

r = 13 cm (check: distances 12 cm and 5 cm, difference 7 ✓)

19* **A regular hexagon is inscribed in a circle of radius r. Find the length of the sides of the hexagon and the distance of each side from the centre of the circle.**

✔ ANSWER

Joining the centre to the 6 vertices gives 6 triangles with central angle 360° ÷ 6 = 60° and two sides r: each is equilateral.

Side = r.

Distance = $\sqrt{(r^2 - (r/2)^2)} = \sqrt{(3r^2/4)} =$ **$(\sqrt{3}/2)\,r$.**

20 **A quadrilateral MNOP is inscribed in a circle. If MN is a diameter, what can you say about ∠MOP and ∠MNP? Explain your reasoning.**

💡 SAMPLE ANSWER

∠MOP and ∠MNP are both subtended by chord MP at points O and N, which lie on the same side of MP (same arc).

So **∠MOP = ∠MNP** (angles in the same segment).

Also, since MN is a diameter, ∠MON = ∠MPN = 90°.

21 **Let ABCD be a cyclic quadrilateral. Explain why the exterior angle at any vertex is equal to the interior opposite angle (e.g., ∠CDE = ∠ABC, where E is on the extension of a side at D).**

✔ ANSWER

Take E on side AD extended beyond D, so ∠CDE is the exterior angle at D.

∠CDE + ∠ADC = 180° (linear pair).

∠ABC + ∠ADC = 180° (opposite angles of a cyclic quadrilateral).

So **∠CDE = ∠ABC**. The same works at every vertex.

22* **"There is no chord of a circle that is longer than its diameter." How do you justify this**

statement?

✔ ANSWER

Let AB be any chord, O the centre.

In ΔOAB (triangle inequality): AB ≤ OA + OB = r + r = 2r = diameter.

Equality only when O is on AB, i.e. AB is a diameter.

Also: the longer chord is closer to the centre and the diameter is at distance 0. **So no chord is longer than the diameter.**

23* **Let A be any point within a circle with centre O. Show that the shortest chord through A is the one perpendicular to OA.**

✔ ANSWER

Take any chord through A; let M be the foot of the perpendicular from O to it. Length = $2\sqrt{(r^2 - OM^2)}$.

In ΔOMA, OM ⊥ MA, so OA is the hypotenuse: OM ≤ OA (equal only when M = A).

So length ≥ $2\sqrt{(r^2 - OA^2)}$, with equality only when M = A, i.e. when the chord is perpendicular to OA at A.

The chord perpendicular to OA is the shortest. (If A = O, all chords through A are diameters.)

24 **How would you use Fig. 5.30 to justify that the angle in a semicircle is 90°?**

(Fig. 5.30: semicircle with centre O; call the ends of the diameter P and Q; A on the arc; OA joined; angles a at P and b at Q.)

✔ ANSWER

OA = OP = OQ (radii), so ΔOAP and ΔOAQ are isosceles.

So ∠OAP = a and ∠OAQ = b.

The angle at A = a + b.

Angle sum of ΔAPQ: a + b + (a + b) = 180° → 2(a + b) = 180°.

a + b = 90°, so **the angle in a semicircle is 90°**.

25* **In a circle, two chords CC' and DD' are drawn perpendicular to a diameter AB. Prove that the segment MM' joining the midpoints of chords CD and C'D' is perpendicular to AB.**

💡 SAMPLE ANSWER

AB passes through the centre and is perpendicular to CC', so it bisects CC' (Theorem 5). So C' is the mirror image of C in line AB. Similarly D' is the mirror image of D.

Using coordinates: take AB as the x-axis. Then C = (p, q), C' = (p, −q), D = (s, t), D' = (s, −t).

M = ((p + s)/2, (q + t)/2), M' = ((p + s)/2, −(q + t)/2).

M and M' have the same x-coordinate, so MM' is a vertical line: **MM' ⊥ AB**.

(If q + t = 0, M and M' coincide on AB.)

26* **How would you use Fig. 5.31 to justify that the sum of the opposite angles of a cyclic quadrilateral is 180°?**

(Fig. 5.31: cyclic quadrilateral ABCD with centre O joined to A, B, C, D.)

💡 **SAMPLE ANSWER**

OA = OB = OC = OD (radii), so ΔOAB, ΔOBC, ΔOCD, ΔODA are isosceles.

Let their equal base angles be w (in ΔOAB), x (ΔOBC), y (ΔOCD), z (ΔODA).

Then ∠A = w + z, ∠B = w + x, ∠C = x + y, ∠D = y + z.

Sum of angles: 2(w + x + y + z) = 360°, so w + x + y + z = 180°.

∠A + ∠C = w + z + x + y = **180°**; similarly ∠B + ∠D = **180°**.

🎉 **Chapter 5 done! Revise the answers once more tomorrow.**